

Pack — Run It Locally

Team setup guide — get the whole stack running on your own machine.

20 June 2026 · branch `tobiloba/engine-spine`

Two levels: **Quick** (infra + engine + frontend → chat with Alpha, create hunts, use Postman) and **Full** (also run the Rust gateway → live hunt canvas/feed).

The `.env` (with the Qwen key) is sent to you separately by Tobi — it's never committed. Drop it at `backend/.env`. Without it the engine still boots in an offline mode (canned replies) so you can test the UI, but Alpha won't be smart.

Prerequisites — install these first

Tool	Version	For	Get it from
Docker Desktop	latest	Postgres + Redis	docker.com
Python	3.12+	the engine	python.org
Node.js	18+	the frontend	nodejs.org
pnpm	latest	frontend packages	<code>npm i -g pnpm</code>
Git	latest	clone the repo	git-scm.com
Rust (cargo)	stable	gateway (live hunts only)	rustup.rs

1 · Get the code

```
git clone <repo-url> pack
cd pack
git checkout tobiloba/engine-spine
```

2 · Start the infra (Postgres + Redis)

```
docker compose up -d      # postgres :5432 and redis :6379
docker compose ps         # both should be "healthy"
```

3 · The engine (Python, port 8000)

```
cd backend
python -m venv .venv
# Windows:      .venv\Scripts\activate
# macOS/Linux: source .venv/bin/activate
pip install -e ".[dev]"

cp .env.example .env      # then replace with the .env Tobi sends (has the Qwen key)

python -m uvicorn app.main:app --port 8000
```

The DB schema is created automatically on boot. Verify: open `http://localhost:8000/health` → `{"status": "ok"}`.

4 • The frontend (React, port 5173)

In a new terminal:

```
cd frontend
pnpm install
cp .env.example .env.local # defaults point at localhost - no secrets here
pnpm dev
```

Open the printed URL (usually `http://localhost:5173`). You can now chat with Alpha and create hunts.

5 • The gateway (Rust, port 8080) — for LIVE hunts

Streams hunt events to the canvas/feed in real time. Skip it if you only need chat/REST.

```
cd gateway
cargo run      # serves the WebSocket on :8080
```

Windows: clone into a path **without spaces** before building the gateway (a Rust/linker bug trips on spaces). The MSVC toolchain needs Visual Studio "Desktop development with C++" build tools.

Ports

Port	Service	Needed for
5432	Postgres (Docker)	always
6379	Redis (Docker)	always
8000	Python engine (REST)	always

5173	Frontend (Vite)	always
8080	Rust gateway (WebSocket)	live hunt streaming

Troubleshooting

- **Alpha gives canned/robotic replies** → no Qwen key; put Tobi's `.env` at `backend/.env` and restart the engine.
- **Canvas never updates during a hunt** → the gateway (`cargo run` in `gateway/`) isn't running.
- **Engine won't start / DB errors** → `docker compose ps` (Postgres healthy?); check `POSTGRES_URL = postgresql://pack:pack@localhost:5432/pack` .
- **Browser calls fail** → engine not on :8000, or `VITE_ENGINE_URL` in `frontend/.env.local` is wrong.

Test with Postman

Import `docs/postman/Pack.postman_collection.json` + the environment, pick **Pack — Local**, and run **Intake** → **Create hunt** → **Approve plan**. Full notes in `docs/postman/README.md` .